

Customer Churn Analysis

PostgreSQL & Python-Based Exploratory and Business Analysis

Database	Analysis	Visualization
PostgreSQL	Python / Pandas	Matplotlib / Seaborn

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1. Executive Summary

This project performs a customer churn analysis using data stored in PostgreSQL and analyzed in Python. Customer, subscription, and support information were cleaned, transformed, combined at customer level, and analyzed to understand churn, retention, revenue, tenure, churn risk, and support activity.

The analysis is descriptive and exploratory. It focuses on identifying patterns within the available customer dataset rather than building a machine-learning churn prediction model.

Metric	Result
Total customers	21
Churned customers	6
Churn rate	28.57%
Retention rate	71.43%
ARPU	18.85
Average tenure	≈ 1,539 days
Revenue at risk	73.94
Escalation rate	19.05%
Average complaints per user	0.43
Escalation–churn correlation	0.77

2. Project Objectives

- Connect Python to a PostgreSQL database.
- Load customer, subscription, and support data into Pandas.
- Explore data quality, structure, missing values, and categories.
- Clean and standardize the source data.
- Engineer churn, complaint, tenure, and churn-risk features.
- Combine the three datasets using customerid.
- Calculate business metrics such as churn, retention, ARPU, and revenue at risk.
- Analyze churn across plans, locations, support activity, and risk categories.
- Create analytical visualizations using Matplotlib and Seaborn.
- Export the processed customer-level dataset to CSV.

3. Technology Stack

Technology	Purpose
Python	Data analysis and preprocessing
Pandas	Data manipulation and transformation

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NumPy	Numerical operations and feature engineering
PostgreSQL	Database storage
psycopg2	PostgreSQL connectivity
SQLAlchemy	Database engine for Pandas
Matplotlib	Visualization
Seaborn	Categorical/statistical visualization
Jupyter Notebook	Analysis environment

4. Dataset and Database Structure

The project uses three PostgreSQL tables connected through the common customerid field.

Table	Main Information
db_customer	Customer profile and demographic information
db_subscription	Subscription, plan, contract, charges, CLTV, churn score
db_support	Complaint dates, escalations, CSAT and support activity

Customer Table

db_customer: customerid, name/customer_name, country, state, gender, dob, interests and pincode. The analysis renamed name to customer_name and removed interests and pincode from the analytical customer data.

Subscription Table

db_subscription: customerid, subscription_start_date, subscription_type, renewal_date, plan_type, contract_type, cancellation_date, cancellation_reason, monthly_charges, cltv and churn_score.

Support Table

db_support: customerid, complaint_date, escalations, csat_score, col_1 and comment. The unused col_1 and comment fields were removed from the analytical data.

5. Data Cleaning and Preparation

Step	Work performed
Empty records	Removed the fully empty customer record.
Column naming	Renamed name to customer_name.
Unused fields	Removed interests, pincode, col_1 and comment from analysis.
Dates	Converted customer, subscription and support date fields to datetime.
Gender	Standardized Men/Women to Male/Female.
Country	Filled missing country values using state-country mapping.
Support records	Sorted support records by complaint date and retained the latest record per customer.

6. Feature Engineering

Churn flag: A customer was marked as churned when cancellation_date was present; otherwise churn_flag was set to 0.

Complaint count: Support records were grouped by customerid to calculate the number of complaint/support records associated with each customer.

Customer tenure: Tenure was calculated from subscription_start_date to cancellation_date for churned customers and from subscription_start_date to the current date for active customers.

Churn risk: The existing churn_score was categorized as low, medium, or high using the thresholds implemented in the notebook. This is a categorization of the supplied score, not a newly trained prediction model.

Cancellation month: A cancellation month field was created to support time-based cancellation analysis.

7. Data Integration

The cleaned subscription data was left-joined with customer information and then with support information using customerid. This produced a customer-level analytical dataset.

The integrated dataset contained 21 customer records before additional analytical transformations.

8. Business Metrics

Churn Rate: 28.57% — 6 of 21 customers were marked as churned.

Retention Rate: 71.43%.

ARPU: 18.85 based on average monthly charges.

Average Tenure: approximately 1,539 days.

Revenue at Risk: 73.94 according to the notebook's calculation, obtained by summing monthly charges for churned customers.

Escalation Rate: 19.05%.

Average Complaints per User: 0.43.

9. Churn by Plan Type

Plan Type	Customers	Monthly Charges	Churn Rate
Basic	5	52.95	60.00%
Premium	7	218.93	14.29%
Standard	9	123.91	22.22%

A pivot-table analysis combined monthly charges, unique customer count, and mean churn_flag to compare plan-level customer and churn metrics.

10. Support and Churn Analysis

Support escalations were converted from Y/N to 1/0 for numerical analysis. The calculated correlation between escalation and churn was **0.77**.

This represents a strong positive linear association within the available dataset. It should not be interpreted as proof that escalations cause churn, particularly because the dataset contains only 21 customers.

11. Exploratory and Visual Analysis

- Churn analysis by state.
- Churn analysis by plan type.
- Monthly charges by plan and gender.
- Churn-risk comparisons.
- Correlation analysis using encoded categorical variables.
- Cancellation-month analysis.
- Customer support and escalation analysis.
- Pivot-table based plan analysis.

12. Final Analytical Dataset

The final analytical dataset combines customer, subscription and support information with engineered features including `churn_flag`, `complaint_count`, `tenure_days`, `churn_risk` and `cancellation_month`.

The processed dataset was exported as **exported_churn_data.csv**.

13. Key Findings

- Overall churn rate was 28.57%, with 6 of 21 customers classified as churned.
- Retention rate was 71.43%.
- The Basic plan had a calculated churn rate of 60.00%, compared with 22.22% for Standard and 14.29% for Premium.
- Premium customers accounted for the largest total monthly charges among the three plans in the analyzed dataset.
- The calculated escalation–churn correlation was 0.77.
- The average calculated customer tenure was approximately 1,539 days.
- Average complaints per user was 0.43.

14. Limitations and Interpretation

- The dataset contains only 21 customer records, so results should be interpreted as descriptive findings for this dataset.
- The existing `churn_score` is supplied by the dataset; the project did not train a new churn prediction model.
- Correlation does not establish causation.
- Revenue at risk is based on the monthly charges of churned customers.
- The analysis is exploratory and is not intended to generalize to a larger customer population without additional data.

15. Skills Demonstrated

PostgreSQL database management and data extraction • Python and Pandas data analysis • NumPy-based feature engineering • Data cleaning and missing-value handling • Date transformation • Categorical standardization • Multi-table data integration • GroupBy and transform operations • Pivot-table analysis • Business metric calculation • Correlation analysis • Matplotlib and Seaborn visualization • CSV data export

16. Conclusion

This project demonstrates a complete customer churn analytics workflow using PostgreSQL and Python. The work progresses from database-connected data extraction through cleaning, feature engineering, customer-level integration, business metric calculation, exploratory analysis, visualization, and final dataset export.

The resulting analysis provides a structured view of customer churn, plan-level behavior, revenue exposure, customer tenure, churn risk, and support activity.